

Data Description and Reproducibility Guide

Project: **Damage Is Not Need** Team: **Auto-City-Research** Project website: ireliya.github.io/auto-city-research Code: github.com/Ireliya/auto-city-research Data: huggingface.co/datasets/Ireliya/auto-city-research

1. Scope and Claim Boundary

This guide documents the data, derived artifacts, fixed configurations, execution order, and verification rules for the submission. The analysis asks where a damage-only post-disaster ranking disagrees with transparent multi-source priority scenarios that additionally represent population exposure, road accessibility, critical facilities, and urban form.

The output is a **priority-disagreement audit**. Neither the multi-source scenarios nor the external proxies are treated as true unmet-need labels, and the workflow is not an operational dispatch system.

The submission contains:

- English research paper: `reports/pdf/paper_en.pdf`;
- Chinese competition report: `reports/pdf/competition_report_cn.pdf`;
- AI collaboration summary and full logs;
- fixed code, configurations, manifests, derived evidence, and figures; and
- a claim-to-evidence ledger at `records/evidence_index.csv`.

2. Dataset Ledger

Source	Role	Spatial/temporal use	Released artifact	Important boundary
xView2/xBD	Building geometry and ordinal damage	Four event footprints; building and 250/500/1,000 m grid aggregation	Derived CSV/GeoJSON and manifests	Raw satellite imagery is not redistributed
WorldPop 100 m R2025A	Primary population exposure	Event-year national rasters intersected with each study grid	Population-enriched grid tables	Population is modeled exposure, not a household census
WorldPop 1 km Global1	Resolution sensitivity	Same events and grids	Population-enriched grid tables	Retained as a lower-resolution comparison
OpenStreetMap current snapshot	Roads, facilities, buildings	Current mapped context for all events	Derived grid metrics and figures	Coverage and post-event edits may differ by place
ohsome API / OSM history	Pre-event roads, facilities, buildings	Snapshot immediately before each event	Event summaries, grid table, fetch log	Failed or low-coverage queries are never converted to zero
CDC SVI 2016	Social vulnerability proxy	Harvey-intersecting Census tracts	Tract aggregates and bootstrap metrics	Measures vulnerability, not realized disaster need
OpenFEMA NFIP Claims v2	Insured-property loss proxy	Harvey-period Texas claims aggregated to tracts	Privacy-safe tract aggregates and bootstrap metrics	Individual claims are never released
OpenFEMA RI-IHP	Assistance proxy	Harvey-relevant ZIP aggregates from one assistance table	ZIP aggregates and bootstrap metrics	Registration and payment processes introduce selection
Census TIGER/Line	Spatial support	Tracts and ZIP-compatible geographies	Only joined aggregates	Source geometries are fetched at acquisition time

References and provider URLs are in `reports/references.bib` and `reports/data_access_license_notes.md`.

3. Study Inventory

Event	xBD buildings	500 m cells	WorldPop year/product
Hurricane Harvey	23,014	612	USA 2017, 100 m primary and 1 km check
Mexico earthquake	32,271	288	Mexico 2017, 100 m primary and 1 km check
Palu tsunami	31,394	196	Indonesia 2018, 100 m primary and 1 km check

Event	xBD buildings	500 m cells	WorldPop year/product
Santa Rosa wildfire	12,950	352	USA 2017, 100 m primary and 1 km check
Total	99,629	1,448	four event footprints

The 100 m population join matched all 1,448 reference cells with no missing, non-finite, or negative values. Event totals and 100 m versus 1 km rank diagnostics are recorded in `data/derived/population_resolution_audit_v1/`.

4. Environment

The published analysis is CPU-only. The reference environment is recorded in `configs/environment_city.yml` and can be checked with:

```
python scripts/verify_city_environment.py
```

Core dependencies include Python 3.11, pandas, NumPy, GeoPandas, Shapely, PyProj, Rasterio, Fiona, Pyogrio, Rtree, Rasterstats, OSMnx, NetworkX, SciPy, scikit-learn, statsmodels, Matplotlib, Seaborn, PyArrow, and PyYAML.

On the reference server, `PROJ_DATA` points to the city environment's `share/proj` directory. `scripts/verify_city_environment.py` tests imports, coordinate transformation, raster support, graph support, and statistical routines.

5. Fixed Analysis Contract

The final audit is governed by `configs/evidence_hardening.yml`, `configs/weight_scenarios.yml`, and `configs/final_evidence.yml`.

- Random seed: 20260715.
- Reference selection budget: exact Top 20%, with $k = \text{ceil}(n \times 0.20)$ per event.
- Damage baselines: area-weighted severity, damage-weighted building area, damaged-building count, and severe/destroyed-building count.
- Weight analysis: 10,000 `Dirichlet(1, 1, 1, 1)` draws and 10,000 policy-bounded draws.
- Policy bounds: damage weight 0.20-0.50; every other component at least 0.05.
- Population products: WorldPop 100 m and 1 km.
- Grid scales: 250 m, 500 m, and 1,000 m, each rebuilt from building-level inputs.
- External-proxy uncertainty: 1,000 fixed-seed bootstrap replicates.

A final robust disagreement must satisfy all non-temporal gates:

- at least three of four damage baselines support the cell;
- policy-bounded disagreement probability is at least 0.80;
- the cell appears under both population products; and
- its area overlaps disagreement evidence at two or more grid scales by at least 0.50.

Historical OSM evidence is reported separately as `support`, `does_not_support`, or `not_assessable`; its threshold is not lowered to increase the number of candidates.

6. Source-to-Result Pipeline

Run acquisition and full reconstruction from the project root in the following order when raw-source access is available:

```
python src/01_parse_xbd_labels.py
python src/02_build_xbd_damage_grid.py
python src/03_fetch_osm_context.py
python src/04_join_worldpop_population.py
python src/05_analyze_priority_mismatch.py
python src/06_profile_mismatch_drivers.py
python src/11_run_robustness_checks.py
python src/15_run_strict_budget_check.py
python src/17_fetch_osm_building_form.py
python src/18_run_baseline_weight_robustness.py
python src/19_run_multiscale_robustness.py
```

```
python src/20_validate_harvey_nfip.py
python src/22_run_historical_osm_sensitivity.py
python src/23_validate_svi_ihp.py
python src/26_audit_population_resolution.py
python src/24_build_final_consensus.py
python src/21_regenerate_publication_figures.py
```

Acquisition scripts expose command-line arguments for paths and output directories. The exact commands used on the reference server are retained in `logs/command_log.md`; public files use portable relative paths.

Main output groups:

Output	Purpose
data/derived/xbd_core_v1/	building-level parsed labels and event inventory
data/derived/worldpop_context_100m_v1/	primary 500 m grid with 100 m population estimates
data/derived/worldpop_context_v1/	1 km population-resolution check
data/derived/priority_mismatch_100m_v1/	primary scenario disagreement
data/derived/evidence_hardening_100m_v1/	damage-baseline and weight uncertainty
data/derived/multiscale_100m_v1/	independently reconstructed 250/500/1,000 m results
data/derived/historical_osm_v1/	pre-event OSM coverage and temporal sensitivity
data/derived/harvey_external_validation_v1/	NFIP tract aggregates and bootstrap metrics
data/derived/external_proxies_v1/	CDC SVI and FEMA assistance proxy metrics
data/derived/final_consensus_v1/	final gate flags, four candidates, event summary, and manifest
reports/figures/	editable SVG, vector PDF, 600 dpi RGB PNG, and grayscale checks

Some internal columns and directories retain earlier `need_*` or `mismatch_*` names so published hashes and schema compatibility are not broken. Their public interpretation is always **multi-source priority scenario** or **priority disagreement**, as fixed in `records/final_terminology_ledger.md`.

7. Public One-Command Reproduction

The public route starts from released derived inputs, so it does not require redistribution of xBD imagery or repeated calls to mutable public APIs:

```
git clone https://github.com/Ireliya/auto-city-research.git
cd auto-city-research
python scripts/reproduce_core.py --profile final
```

The command performs six checks:

- download the Hugging Face dataset revision fixed in `configs/public_release.yaml`;
- verify file sizes and SHA-256 hashes from `MANIFEST.csv`;
- run data and schema smoke tests;
- recompute the legacy 1 km core and verify the published 67/109 invariants;
- recompute the 100 m analysis, weight robustness, multiscale audit, and final consensus; and
- verify the exact four released candidate keys and regenerate Figures 11-12.

For a local dataset already downloaded from the fixed revision:

```
python scripts/reproduce_core.py --profile final --skip-download
```

The reproduction report is written to `data/reproduced/core_v1/reproduction_report.json`. Time fields may differ; deterministic numerical outputs and candidate keys must not.

8. Verification Invariants

The final package and continuous-integration checks enforce:

- strict Top-k size equals `ceil(n x share)` for every event and threshold;
- all analysis metrics are finite;
- primary 100 m provisional counts are 73 by percentile consensus and 115 by exact Top 20%;

- legacy 1 km checks remain 67 and 109;
- the final event counts are Harvey 0, Mexico 4, Palu 0, Santa Rosa 0;
- no final candidate is labeled temporally supported;
- figure exports exist in all four required formats; and
- no raw imagery, raw claim records, credentials, private paths, or cache files enter the public package.

9. Reproducibility Levels

Level	What can be reproduced	Requirements
Public deterministic	Core rankings, robustness tables, final consensus, final evidence figures	GitHub code + pinned Hugging Face derived data
Full source reconstruction	xBD parsing, WorldPop joins, OSM acquisition, external proxy aggregation	Original source access and live public endpoints
Submission audit	PDFs, package manifest, hashes, path/secret checks	Complete competition project directory

Network-dependent outputs are released with fetch logs and manifests. A failed API request is an error or `not_assessable`, never a zero observation.

10. Data Protection and Redistribution

Included:

- derived and aggregated tables;
- selected derived geospatial layers;
- scripts, fixed configs, logs, manifests, and checksums;
- reports and figures.

Excluded:

- raw xBD satellite imagery and full source archive;
- raw WorldPop rasters and bulk OSM extracts/caches;
- individual NFIP or FEMA assistance records;
- credentials, IP addresses, private absolute paths, and temporary files.

The public artifact supports scientific audit but does not replace provider-specific source access. See `reports/data_access_license_notes.md` for attribution and source-level restrictions.

11. Known Validity Boundaries

- No observed variable is a ground-truth label for unmet rescue or recovery need.
- The four retained cells are robust to the fixed non-temporal specifications but are not supported by historical OSM persistence.
- OSM completeness varies spatially and historically; one Harvey historical comparison is `not_assessable` under the fixed coverage rule.
- WorldPop is model-based and resolution-sensitive; NFIP, SVI, and FEMA assistance measure distinct selected populations and processes.
- The selected xBD footprints are not complete disaster extents, and the audit is neither causal evidence nor an autonomous allocation rule.